Assignment #4 - Matthew Athanasopoulos (20976490)

CRANK-NICHOLSON CHOSEN!

Problem: $\dfrac{\partial^2 u}{\partial r^2} + \dfrac{1}{r}\dfrac{\partial u}{\partial r} = \dfrac{\partial u}{\partial z}$, $u(1,z)=1$, $\dfrac{\partial u}{\partial r}(0,z)=0$, Initial condition

treating the axial direction like time, $z \to t$

$\partial u/\partial r = 0$ $\quad$ $u=1$

$i=0$ $\quad i=1$ $\quad i=2$ $\quad i=3$ $\quad i=4$ $\quad i=5$

$r=0$ $\quad r=0.2$ $\quad r=0.4$ $\quad r=0.6$ $\quad r=0.8$ $\quad r=1$

$\longrightarrow r$

Finite Difference: (letting $z \to t$)

$$\frac{\partial^2 u}{\partial r^2} = \frac{1}{2}\left(\frac{u_{i+1}^{l} - 2u_i^{l} + u_{i-1}^{l}}{(\Delta r)^2} + \frac{u_{i+1}^{l+1} - 2u_i^{l+1} + u_{i-1}^{l+1}}{(\Delta r)^2}\right)$$

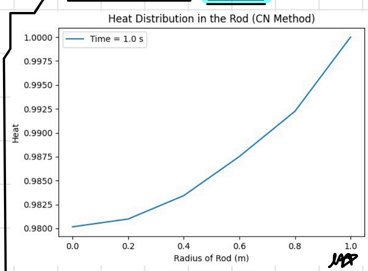
$$\frac{\partial u}{\partial r} = \frac{1}{2}\left(\frac{u_{i+1}^{l} - u_{i-1}^{l}}{2\Delta r} + \frac{u_{i+1}^{l+1} - u_{i-1}^{l+1}}{2\Delta r}\right)$$

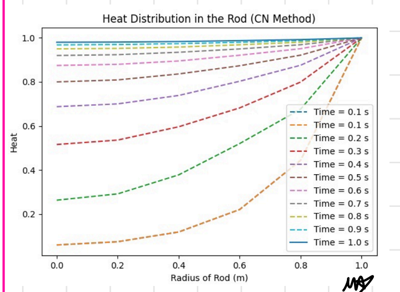
$$\frac{\partial u}{\partial t}: \quad \frac{u_i^{l+1} - u_i^{l}}{(\frac{1}{2})2\Delta t} = \frac{u_i^{l+1} - u_i^{l}}{\Delta t}$$

Putting into matrix form:

$$\begin{bmatrix} 3 & -4 & 1 & 0 & 0 \\ -\lambda\left(1-\frac{\Delta r}{2r_1}\right) & 1+2\lambda & -\lambda\left(1+\frac{\Delta r}{2r_1}\right) & 0 & 0 \\ 0 & -\lambda\left(1-\frac{\Delta r}{2r_1}\right) & 1+2\lambda & -\lambda\left(1+\frac{\Delta r}{2r_1}\right) & 0 \\ 0 & 0 & -\lambda\left(1-\frac{\Delta r}{2r_1}\right) & 1+2\lambda & -\lambda\left(1+\frac{\Delta r}{2r_1}\right) \\ 0 & 0 & 0 & -\lambda\left(1-\frac{\Delta r}{2r_1}\right) & 1+2\lambda \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} -u_2^l + 4u_1^l - 3u_0^l \\ \lambda\left(1+\frac{\Delta r}{2r_1}\right)u_2^l + (1-2\lambda)u_1^l + \lambda\left(1-\frac{\Delta r}{2r_1}\right)u_0^l \\ \lambda\left(1+\frac{\Delta r}{2r_2}\right)u^l + (1-2\lambda)u_2^l + \lambda\left(1-\frac{\Delta r}{2r_2}\right)u_1^l \\ \lambda\left(1+\frac{\Delta r}{2r_3}\right)u^l + (1-2\lambda)u_3^l + \lambda\left(1-\frac{\Delta r}{2r_3}\right)u_2^l \\ 2\lambda\left(1+\frac{\Delta r}{2r_4}\right) + (1-2\lambda)u_4^l + \lambda\left(1-\frac{\Delta r}{2r_4}\right)u_3^l \end{bmatrix}$$

CRANK-NICHOLSON **PLOTS:**





Substituting into the original equation:

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} = \frac{\partial u}{\partial z} \qquad \text{Sub } z \to t$$

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} = \frac{\partial u}{\partial t}$$

$$\frac{1}{2}\left(\frac{u_{i+1}^{l} - 2u_i^{l} + u_{i-1}^{l}}{(\Delta r)^2} + \frac{u_{i+1}^{l+1} - 2u_i^{l+1} + u_{i-1}^{l+1}}{(\Delta r)^2}\right) + \frac{1}{r}\cdot\frac{1}{2}\left(\frac{u_{i+1}^{l} - u_{i-1}^{l}}{2\Delta r} + \frac{u_{i+1}^{l+1} - u_{i-1}^{l+1}}{2\Delta r}\right) = \frac{u_i^{l+1} - u_i^{l}}{\Delta t}$$

rearranging

$$\frac{\Delta t}{2(\Delta r)^2}\left(u_{i+1}^{l} - 2u_i^{l} + u_{i-1}^{l} + u_{i+1}^{l+1} - 2u_i^{l+1} + u_{i-1}^{l+1}\right) + \frac{\Delta t}{4r\Delta r}\left(u_{i+1}^{l} - u_{i-1}^{l} + u_{i+1}^{l+1} - u_{i-1}^{l+1}\right) = u_i^{l+1} - u_i^{l}$$

Let $\lambda = \dfrac{\Delta t}{2(\Delta r)^2}$

$$\lambda\left(u_{i+1}^{l} - 2u_i^{l} + u_{i-1}^{l} + u_{i+1}^{l+1} - 2u_i^{l+1} + u_{i-1}^{l+1}\right) + \lambda\frac{\Delta r}{2r}\left(u_{i+1}^{l} - u_{i-1}^{l} + u_{i+1}^{l+1} - u_{i-1}^{l+1}\right) = u_i^{l+1} - u_i^{l}$$

Expanding and grouping l and $l+1$ terms

$$-\lambda\left(1+\frac{\Delta r}{2r_i}\right)u_{i+1}^{l+1} + (1+2\lambda)u_i^{l+1} - \lambda\left(1-\frac{\Delta r}{2r_i}\right)u_{i-1}^{l+1} = \lambda\left(1+\frac{\Delta r}{2r_i}\right)u_{i+1}^{l} + (1-2\lambda)u_i^{l} + \lambda\left(1-\frac{\Delta r}{2r_i}\right)u_{i-1}^{l}$$

FOR A GENERIC NODE "i".

Apply B.C's

$u(1,t)=1$ is refering to the last node, $u_5^l = 1$ for the setup

$$\frac{\partial u}{\partial t} = 0: \ \frac{1}{2}\left(\frac{u_{i+1}^{l} - u_{i-1}^{l}}{2\Delta r} + \frac{u_{i+1}^{l+1} - u_{i-1}^{l+1}}{2\Delta r}\right) = 0 \Rightarrow -u_{i+2}^{l} + 4u_{i+1}^{l} - 3u_i^{l} - u_{i+2}^{l+1} + 4u_{i+1}^{l+1} - 3u_i^{l+1} = 0$$

$$u_{i+2}^{l+1} - 4u_{i+1}^{l+1} + 3u_i^{l+1} = -u_{i+2}^{l} + 4u_{i+1}^{l} - 3u_i^{l}$$

FIRST NODE

For each node:

$i=0$: $u_2^{l+1} - 4u_1^{l+1} + 3u_0^{l+1} = -u_2^l + 4u_1^l - 3u_0^l$

$i=1$: $-\lambda\left(1+\frac{\Delta r}{2r_1}\right)u_2^{l+1} + (1+2\lambda)u_1^{l+1} - \lambda\left(1-\frac{\Delta r}{2r_1}\right)u_0^{l+1} = \lambda\left(1+\frac{\Delta r}{2r_1}\right)u_2^l + (1-2\lambda)u_1^l + \lambda\left(1-\frac{\Delta r}{2r_1}\right)u_0^l$

$i=2$: $-\lambda\left(1+\frac{\Delta r}{2r_2}\right)u_3^{l+1} + (1+2\lambda)u_2^{l+1} - \lambda\left(1-\frac{\Delta r}{2r_2}\right)u_1^{l+1} = \lambda\left(1+\frac{\Delta r}{2r_2}\right)u_3^l + (1-2\lambda)u_2^l + \lambda\left(1-\frac{\Delta r}{2r_2}\right)u_1^l$

$i=3$: $-\lambda\left(1+\frac{\Delta r}{2r_3}\right)u_4^{l+1} + (1+2\lambda)u_3^{l+1} - \lambda\left(1-\frac{\Delta r}{2r_3}\right)u_2^{l+1} = \lambda\left(1+\frac{\Delta r}{2r_3}\right)u_4^l + (1-2\lambda)u_3^l + \lambda\left(1-\frac{\Delta r}{2r_3}\right)u_2^l$

$i=4$: $-\lambda\left(1+\frac{\Delta r}{2r_4}\right)\underbrace{\cancel{u_5^{l+1}}}_{=1\,(BC)} + (1+2\lambda)u_4^{l+1} - \lambda\left(1-\frac{\Delta r}{2r_4}\right)u_3^{l+1} = \lambda\left(1+\frac{\Delta r}{2r_4}\right)\underbrace{\cancel{u_5^l}}_{=1\,(BC)} + (1-2\lambda)u_4^l + \lambda\left(1-\frac{\Delta r}{2r_4}\right)u_3^l$

$\cancel{+\lambda\left(1+\frac{\Delta r}{2r_4}\right)} \cancel{-\lambda\left(1+\frac{\Delta r}{2r_4}\right)} + (1+2\lambda)u_4^{l+1} - \lambda\left(1-\frac{\Delta r}{2r_4}\right)u_3^{l+1} = \lambda\left(1+\frac{\Delta r}{2r_4}\right) + (1-2\lambda)u_4^l + \lambda\left(1-\frac{\Delta r}{2r_4}\right)u_3^l + \lambda\left(1+\frac{\Delta r}{2r_4}\right)$

$(1+2\lambda)u_4^{l+1} - \lambda\left(1-\frac{\Delta r}{2r_4}\right)u_3^{l+1} = 2\lambda\left(1+\frac{\Delta r}{2r_4}\right) + (1-2\lambda)u_4^l + \lambda\left(1-\frac{\Delta r}{2r_4}\right)u_3^l$

$i=5$: $u(1,t)=1 \Rightarrow u_5^{l+1} = u_5^l = 1$

FEA:

$$\frac{\partial^2 T}{\partial x^2} = -f(x), \quad f(x) = 15, \quad T(x=0) = 75, \quad T(x=L=10) = 150$$

$$\int \frac{\partial^2 T}{\partial x^2} = \int -15$$

$$\int \frac{\partial T}{\partial x} = \int -15x - C_1$$

$$T = -\frac{15}{2}x^2 + C_1 x + C_2$$

Apply B.C's

$T(x=0) = 75$

$$T = \frac{-15}{2}x^2 + c_1 x + C_2$$

$$75 = \frac{-15}{2}(0)^2 + c_1(0) + C_2 \Rightarrow C_2 = 75$$

$T(x=L=10) = 150$

$$T = \frac{-15}{2}x^2 + c_1 x + 75$$

$$150 = \frac{-15}{2}(10)^2 + c_1(10) + 75$$

$$150 = -750 + c_1(10) + 75 \Rightarrow c_1 = 82.5$$

$$\Rightarrow T = \frac{-15}{2}x^2 + 82.5x + 75$$

For the FEA, we have 4 elements (5 nodes):

$$\partial x = \frac{10}{4} = 2.5$$

$$\overset{1}{\bullet}\underbrace{\quad}_{1}\overset{2}{\bullet}\underbrace{\quad}_{2}\overset{3}{\bullet}\underbrace{\quad}_{3}\overset{4}{\bullet}\underbrace{\quad}_{4}\overset{5}{\bullet}$$

Node 1 = $N_1 = \frac{x_2 - x}{x_2 - x_1}$ Node 2 = $N_2 = \frac{x - x_1}{x_2 - x_1}$ Element 1: $T_1 = N_1 T_1 + N_2 T_2$

By weighted residuals, we can attempt to minimize R through Galerkin's method.

$$\frac{\partial^2 T}{\partial x^2} = -15 \Rightarrow \frac{\partial^2 T}{\partial x^2} + 15 = R \longrightarrow \int_{x_1}^{x_2} \left(\frac{\partial^2 T}{\partial x^2} + 15\right) N_i \, dx = -\int_{x_1}^{x_2} 15(N_i)\, dx$$

$$\int_{x_1}^{x_2} \frac{\partial^2 T}{\partial x^2} N_i \, dx = -\int_{x_1}^{x_2} f(x) N_i \, dx = N_i \frac{\partial T}{\partial x}\Big|_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{\partial T}{\partial x}\frac{\partial N_i}{\partial x}\, dx$$

$$N_i \frac{\partial T}{\partial x}\Big|_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{\partial T}{\partial x}\frac{\partial N_i}{\partial x}\, dx = -\int_{x_1}^{x_2} f(x) N_i \, dx$$

$$\int_{x_1}^{x_2} \frac{\partial T}{\partial x}\frac{\partial N_i}{\partial x}\, dx = N_i \frac{\partial T}{\partial x}\Big|_{x_1}^{x_2} + \int_{x_1}^{x_2} f(x) N_i \, dx$$

$$\int_{x_1}^{x_2} \frac{\partial T}{\partial x}\frac{\partial N_1}{\partial x}\, dx = N_1 \frac{\partial T}{\partial x}\Big|_{x_1}^{x_2} + \int_{x_1}^{x_2} f(x) N_1 \, dx \quad \text{NODE 1 } (i=1)$$

$$\int_{x_1}^{x_2} \frac{\partial T}{\partial x}\frac{\partial N_2}{\partial x}\, dx = N_2 \frac{\partial T}{\partial x}\Big|_{x_1}^{x_2} + \int_{x_1}^{x_2} f(x) N_2 \, dx \quad \text{NODE 2 } (i=2)$$

Internal system | B.C's | External Force

$$T = N_1 T_1 + N_2 T_2$$

$$\frac{\partial T}{\partial x} = T_1 \frac{\partial N_1}{\partial x} + T_2 \frac{\partial N_2}{\partial x} \Rightarrow N_1 = \frac{x_2 - x}{x_2 - x_1} \Rightarrow \frac{\partial N_1}{\partial x} = \frac{-1}{x_2 - x_1}$$

$$\frac{\partial T}{\partial x} = T_1\left(\frac{-1}{x_2 - x_1}\right) + T_2\left(\frac{1}{x_2 - x_1}\right) \Rightarrow N_2 = \frac{x - x_1}{x_2 - x_1} \Rightarrow \frac{\partial N_2}{\partial x} = \frac{1}{x_2 - x_1}$$

$$\Rightarrow \frac{\partial T}{\partial x} = \frac{T_2 - T_1}{x_2 - x_1}$$

For internal system:

Node 1: $$\int_{x_1}^{x_2} \frac{\partial T}{\partial x}\frac{\partial N_1}{\partial x}\, dx = \int_{x_1}^{x_2} \frac{T_2 - T_1}{x_2 - x_1}\left(\frac{-1}{x_2 - x_1}\right) dx = -\frac{T_2 - T_1}{(x_2 - x_1)^{\cancel{2}}}\cancel{(x_2 - x_1)} = -\frac{T_2 - T_1}{x_2 - x_1}$$

SAME APROACH DIFF SIGN.

Node 2: $$\int_{x_1}^{x_2} \frac{\partial T}{\partial x}\frac{\partial N_2}{\partial x}\, dx = \cdots = \frac{T_2 - T_1}{x_2 - x_1}$$

For B.C's

Node 1: $$\int_{x_1}^{x_2} \frac{\partial T}{\partial x}\frac{\partial N_1}{\partial x}\, dx = N_1 \frac{\partial T}{\partial x}\Big|_{x_1}^{x_2} + \int_{x_1}^{x_2} f(x) N_1\, dx$$

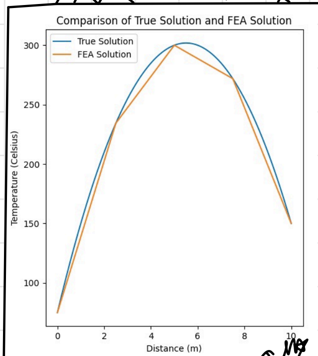
Node 2: $$\int_{x_1}^{x_2} \frac{\partial T}{\partial x}\frac{\partial N_2}{\partial x}\, dx = N_2 \frac{\partial T}{\partial x}\Big|_{x_1}^{x_2} + \int_{x_1}^{x_2} f(x) N_2\, dx$$

$$N_1(x_2) = \frac{\cancel{x_2 - x_2}}{x_2 - x_1} \overset{0}{=} 0, \quad N_1(x_1) = \frac{x_2 - x_1}{x_2 - x_1} = 1$$

$$\Rightarrow N_1 \frac{\partial T}{\partial x}\Big|_{x_1}^{x_2} = \frac{-\partial T}{\partial x}(x_1)$$

By the same approach (except 1 and 0 will flip for $x = x_1$ and $x = x_2$...

$$N_2 \frac{\partial T}{\partial x}\Big|_{x_1}^{x_2} = \frac{\partial T}{\partial x}(x_2)$$



By np.linalg.solve:

$T_1 = 75$, $T_2 = 234.375$, $T_3 = 300$

$T_4 = 271.875$, $T_5 = 150$

For External Force:

Node 1:

$$\int_{x_1}^{x_2} \frac{\partial T}{\partial x}\frac{\partial N_1}{\partial x}\, dx = N_1 \frac{\partial T}{\partial x}\Big|_{x_1}^{x_2} + \int_{x_1}^{x_2} f(x) N_1\, dx$$

$$\int_{x_1}^{x_2} f(x) N_1\, dx = \int_{x_1}^{x_2} 15 \cdot N_1\, dx \quad x_1 = 0, \; x_2 = 2.5$$

$$\Rightarrow 15\int_0^{2.5}\left(1 - \frac{x}{2.5}\, dx\right) = 15\left(x\Big|_0^{2.5} - \frac{x^2}{5}\Big|_0^{2.5}\right) = 18.75$$

Node 2.

$$\int_{x_1}^{x_2} \frac{\partial T}{\partial x}\frac{\partial N_2}{\partial x}\, dx = N_2 \frac{\partial T}{\partial x}\Big|_{x_1}^{x_2} + \int_{x_1}^{x_2} f(x) N_2\, dx$$

$$\int_{x_1}^{x_2} f(x) N_2\, dx \quad x_1 = 0, \; x_2 = 2.5$$

$$\int_{x_1}^{x_2} 15 \cdot N_2\, dx = 15\int_0^{2.5} \frac{x}{2.5}\, dx = 15\left(\frac{x^2}{5}\Big|_0^{2.5}\right)$$

$$\int_{x_1}^{x_2} f(x) N_2\, dx = 18.75$$

Plugging into original Equation:

Node 1: $\frac{T_1 - T_2}{x_2 - x_1} = -\frac{\partial T}{\partial x}(x_1) + 18.75$: $x_1 = 0$, $x_2 = 2.5$

$$\Rightarrow 0.4T_1 - 0.4T_2 = \frac{-\partial T}{\partial x}(x_1) + 18.75$$

Node 2: $-\frac{T_1 - T_2}{x_2 - x_1} = \frac{\partial T}{\partial x}(x_2) + 18.75$: $x_1 = 0$, $x_2 = 2.5$

$$\Rightarrow -0.4T_1 + 0.4T_2 = \frac{\partial T}{\partial x}(x_2) + 18.75$$

*ELEMENT 2,3,4 SHARE NODES, WE CAN ADD TO GET RID OF DERIVATIVES THIS WAY.

Element #1

NODE 1 $\Rightarrow 0.4T_1 - 0.4T_2 = \frac{-\partial T}{\partial x}(x_1) + 18.75$

NODE 2 $\Rightarrow -0.4T_1 + 0.4T_2 = \frac{\partial T}{\partial x}(x_2) + 18.75$

Element #2

NODE 2 $\Rightarrow 0.4T_2 - 0.4T_3 = \frac{-\partial T}{\partial x}(x_2) + 18.75$

NODE 3 $\Rightarrow -0.4T_2 + 0.4T_3 = \frac{\partial T}{\partial x}(x_3) + 18.75$

Element #3:

NODE 3 $\Rightarrow 0.4T_3 - 0.4T_4 = \frac{-\partial T}{\partial x}(x_3) + 18.75$

NODE 4 $\Rightarrow -0.4T_3 + 0.4T_4 = \frac{\partial T}{\partial x}(x_4) + 18.75$

Element #4:

NODE 4 $\Rightarrow 0.4T_4 - 0.4T_5 = \frac{-\partial T}{\partial x}(x_4) + 18.75$

NODE 5 $\Rightarrow -0.4T_4 + 0.4T_5 = \frac{\partial T}{\partial x}(x_5) + 18.75$

By adding the like equations $(x_i + x_i)$:

$\Rightarrow$ Node 2: $-0.4T_1 + 0.8T_2 - 0.4T_3 = 37.5$

$\Rightarrow$ Node 3: $-0.4T_2 + 0.8T_3 - 0.4T_4 = 37.5$

$\Rightarrow$ Node 4: $-0.4T_3 + 0.9T_4 - 0.4T_5 = 37.5$

Since $T_1 = 75$ and $T_5 = 150$:

$\Rightarrow$ Node 1: $\frac{\partial T}{\partial x}(x_1) = 0.4T_2 - 11.25$

$\Rightarrow$ Node 2: $0.8T_2 - 0.4T_3 = 67.5$

$\Rightarrow$ Node 3: $-0.4T_2 + 0.8T_3 - 0.4T_4 = 37.5$

$\Rightarrow$ Node 4: $-0.4T_3 + 0.8T_4 = 97.5$

$\Rightarrow$ Node 5: $\frac{\partial T}{\partial x} = -0.4T_4 + 41.25$

MATRIX FORM:

$$\begin{bmatrix} 0.8 & -0.4 & 0 \\ -0.4 & 0.8 & -0.4 \\ 0 & -0.4 & 0.8 \end{bmatrix}\begin{bmatrix} T_2 \\ T_3 \\ T_4 \end{bmatrix} = \begin{bmatrix} 67.5 \\ 37.5 \\ 97.5 \end{bmatrix}$$

$\frac{\partial T}{\partial x}(x_1) = 0.4T_2 - 11.25$

$\frac{\partial T}{\partial x}(x_5) = -0.4T_4 + 41.25$

We've obtained the system of equations!!!

Analytical Solution: $T = \frac{-15}{2}x^2 + 82.5x + 75$

$\rightarrow$ For comparison with python FEA solution!

.py file attached!!